

new visual cognition algorithms towards human-level visual concept learning and reasoning. We envision our benchmark to serve as a driving force for research on context-dependent and analogical perception beyond standard visual recognition. We believe that endowing machine perception with the abilities to learn and reason in a human-like way is an essential step towards building robust and reliable AI systems in the wild. It could potentially lead to more human-interpretable AI systems and address concerns about ethics and fairness arising from today’s data-driven learning systems that inherit or augment the biases in training data. A potential risk of our new benchmark is that it might skew research towards highly customized methods without much applicability for more general concept learning and reasoning. We encourage researchers to develop new algorithms for our benchmark from the first principles and avoid highly customized solutions, to retain the generality and broad applicability of the resultant algorithms.

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